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Prediction of Moisture Content in Gambir Leaf Herbal Tea Using NIR Spectroscopy

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Abstract. Gambir leaf herbal tea is an innovative product made from Gambir (*Uncaria gambir* Roxb.), which can potentially be an export product. Considering its hygroscopic nature, moisture content is an imperative indicator for controlling the stability of Gambir leaf herbal tea quality. This research intends to investigate near-infrared spectroscopy (NIR) to estimate the moisture content of herbal tea made from Gambir leaves. We placed powdered Gambir leaf herbal tea samples in a petri dish, using 20–21 g per sample. The number of samples used was 60. We obtained the reflectance of herbal tea from Gambir leaves using NIR spectroscopy at 1000 to 2500 nm. The oven method determined the reference moisture content. A calibration model was developed using the partial least squares (PLS) regression method. Furthermore, the model acquires standard error in the calibration (SEC) and prediction (SEP) are 0.36 and 0.43, respectively. The model had a determination coefficient of prediction set (R^2p) of 0.95 and Residual Predictive Deviations (RPD) of 4.38. This study proves that NIR spectroscopy is reliable in presupposing the moisture content of herbal tea made from Gambir leaves.

Keywords: Gambir leaves; Herbal tea; Moisture content; NIR spectroscopy; PLS

1. Introduction

Gambir leaf herbal tea (*Uncaria gambir* Roxb.) has health benefits because it contains polyphenol compounds that are good for the body and function as antioxidants and antibacterials [1]. Indonesia controls around 80% of the world's Gambir market, but most of it is exported as crude extracts, not finished products. This shows an opportunity to increase added value by producing herbal tea from Gambir leaves as a more valuable export product [2]. Product quality is a key factor in entering the global market. One important parameter in maintaining product quality is measuring moisture content to minimize damage to the material [3]. Gambir leaf powder is hygroscopic, quickly absorbing water vapor from the environment. High moisture content can trigger the product to become damp and easily damaged [4]. Therefore, controlling moisture content is very important.

Traditional methods such as oven drying and Karl Fischer titration are time-consuming and destructive and are not ideal for fast industrial needs. Near Infrared (NIR) spectroscopy technology offers a non-destructive solution quickly and can provide real-time output information with good measurement results [5]. The Partial Least Squares (PLS) method was utilized to investigate the relationship between the moisture content of herbal tea derived from

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Gambir leaves and the NIR spectrum. We chose this method because it can process large amounts of independent variable data, enabling fast and efficient data collection and preparation processes and fast and environmentally friendly detection processes [6]. This research intends to assess the use of NIR spectroscopy in forecasting moisture content in herbal tea made from Gambir leaves. Furthermore, NIR spectroscopy technology is expected to be an efficient quality control method, especially for export products, thereby increasing competitiveness in the global market and extending the product's shelf life.

2. Materials and methods

2.1 Sample preparation

Gambir leaves of the Cubadak variety were obtained from Nagari Siguntur, West Sumatra. Before the drying process, the Gambir leaves were first washed and then aired at room temperature of 20-26°C for 17 hours to reduce the moisture content [7]. Furthermore, the drying process was carried out using a rack-type oven at 45°C [8], with drying duration of 6, 8, and 10 hours. The desiccated leaves were subsequently minced using a chopper and sifted through a 40-mesh sieve [9], to get herbal tea powder. This investigation utilized 60 samples, each weighing between 20 and 21 grams.

2.2 NIR spectrum acquisition for herbal tea from Gambir leaves

NIR spectrum acquisition for Gambir leaf herbal tea using Buchi NIRFlex N-500 Solids instrument. Each sample was placed into a petri dish with a diameter of 10 cm, then scanned at a wave number of 10 000 – 4 000 cm^{-1} or a wavelength of 1 000 – 2 500 nm. Reflectance spectrum acquisition was carried out three times for each sample.

2.3 Measurement of moisture content

The measurement of moisture content refers to the Indonesian National Standard (SNI), which stipulates that the maximum moisture content for herbal tea powder is no more than 8% so that the quality and stability of the product are maintained during storage [10]. The process begins with sterilizing the cup using an oven at a 105°C for 15 minutes. Next, the cup is filled with 5 grams of gambir leaf powder sample and left for 3 hours in the oven; then, the sample is subjected to cooling in a desiccator for three minutes, followed by weighing. The drying process is stopped if the sample weight is constant for three consecutive measurements. The results of this analysis are then reference data. The moisture content is quantified using the subsequent equation:

$$\text{Moisture content (MC)} = \frac{b-c}{c-a} \times 100\% \quad (1)$$

Where:
MC = Moisture Content (%)
a = Weight of cup (gram)
b = Weight of cup + sample before drying (gram)
c = Weight of cup + sample after drying (gram)

2.4 Moisture content prediction model for herbal tea from Gambir leaves

The reflectance spectrum that has been collected is then converted into an absorbance spectrum (Equation 2) using the MS. Excel application [11]. The prediction model is built using the original NIR spectrum as the X variable and the reference moisture content as the Y variable using The Unscrambler X 10.3 application. This process begins with building a calibration model using 2/3

of the total sample, then followed by a validation process to test the model. The calibration model is quantified and validated using the cross-validation method with both internal and external samples (1/3 of the total sample). The model is constructed using the Partial Least Squares (PLS) regression. The performance of the calibration model is based on several statistical parameters, namely the coefficient of determination (R^2), Residual Predictive Deviations (RPD), Standard Error of Calibration (SEC) and Standard Error of Prediction (SEP).

$$Absorbance = Log\left(\frac{1}{R}\right) \quad (2)$$

3. Results and discussion

3.1 Statistic of moisture content of herbal tea from Gambir leaves

Table 1 shows the statistical data of the moisture content of herbal tea from Gambir leaves in the calibration and validation sets. The moisture content actual data used to build the model (calibration set) has a more expansive data range than the data for testing the model (validation set). Consequently, the actual moisture content data in the validation set falls within the range of the calibration set values. [12]. The model developed must have a broad data distribution so that the data in the validation set can assess the performance of the calibrated model [13].

Table 1. Moisture content of Gambir leaf herbal tea statistic in calibration and validation set

Sample Set	n	Max.	Min.	Mean	S.D.
Calibration Set	40	15.90	4.72	8.85	2.43
Validation Set	20	10.82	5.21	7.80	1.81

n = number of samples; Max. = Maximum; Min = Minimum; S.D. = Standard Deviation

3.2 Spectra investigation

Figure 1 shows the raw spectra of the NIR spectrum of herbal tea from Gambir leaves. The spectrum displays peaks indicating the presence of certain chemical bonds. NIR waves that interact with chemical bonds in the sample will produce vibrations so that peaks of waves are visible in the NIR spectrum. This study found that the resulting spectrum indicates the presence of chemical compounds relevant to the moisture content in herbal tea made from Gambir leaves. Water content is generally detected in the waveform range of around 1400–1450 nm and 1900–1950 nm [14]. This range is related to water absorption due to water molecules' vibration in the O-H bond [15].

The PLS method was applied to extract meaningful information from the NIR spectrum while separating relevant signals from noise [16]. This study demonstrates that the PLS approach effectively estimates the moisture content of herbal tea derived from Gambir leaves. The scatter plot graph (Figure 2) illustrates the prediction results, demonstrating a strong correlation between the reference and NIR prediction values. Moisture content predicts NIR is very close to the predetermined reference value, which indicates that the PLS method has a high ability to

forecast the moisture content of herbal tea made by Gambir leaves with a satisfactory level of accuracy [17][18][19].

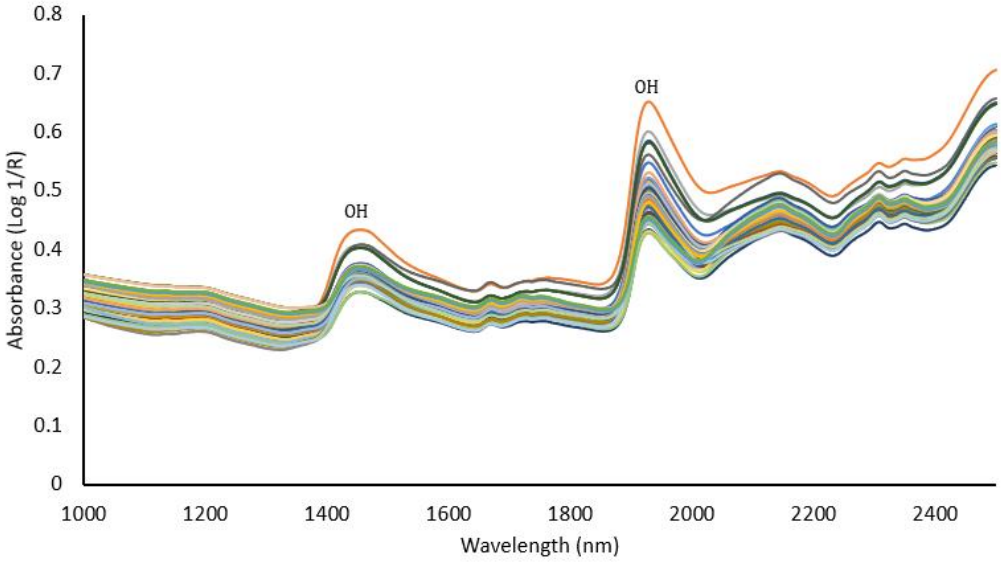


Figure 1. Raw spectra of herbal tea from Gambir leaves.

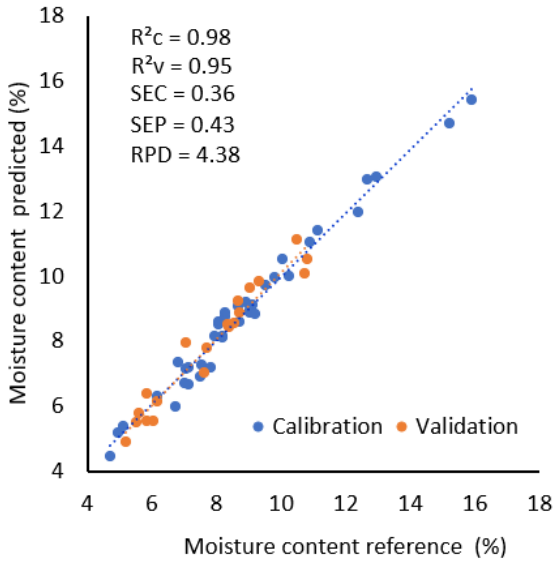


Figure 2. Scatter plot of moisture content from reference measurement and NIR prediction.

Furthermore, the resulting model has a coefficient of determination value on the calibration set (R^2c) and the validation set (R^2v) of 0.98 and 0.95, respectively, illustrating the model's ability to explain data variation. In addition, the SEC value of 0.36 and SEP of 0.43 indicate a high level of model precision in predicting moisture content. A accurate calibration model demonstrates a minor discrepancy between SEC and SEP values [20], [21]. The RPD value of 4.38 means the model built has excellent predictive ability. According to the literature, a NIR model can make perfect

predictions if it has an RPD value > 3[22], and large data diversity will increase the RPD value [23]. Thus, this study has proven that the NIR spectroscopy and PLS methods are reliable in forecasting the moisture content of herbal tea made from Gambir leaves and provide valuable insights into the development of herbal tea products.

4. Conclusion

The conducted research and analysis demonstrate that the integration of NIR spectroscopy and PLS is highly effective for estimating moisture content in herbal tea derived from Gambir leaves. The developed prediction model has been shown to be effective in forecasting the moisture content of herbal tea derived from Gambir leaves, as evidenced by statistical parameters such as R^2c of 0.98, R^2v of 0.95, SEC of 0.36, SEP of 0.43 and RPD value of 4.38.

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